



PUBLIC SMART CONTRACT AUDIT REPORT

for BiFi Protocol

BiFi Lending Protocol -- Cross-Chain Bridge

Prepared By: Red Ghost Security

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Version	Date	Author(s)	Description
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1. Introduction

1.1 About the Audited Protocol

Red Ghost Security was engaged to perform a security audit of the BiFi Protocol protocol, a Cross-Chain Lending Protocol operating on Multi-Chain (Ethereum / Polygon / BNB Chain). The audit was conducted between May 8, 2026 and May 9, 2026.

The BiFi Protocol protocol enables users to participate in cross-chain lending protocol with a focus on capital efficiency and composability. The protocol's smart contracts were reviewed for security vulnerabilities, logical inconsistencies, and adherence to smart contract security best practices.

The audit methodology included manual code review, Foundry-based invariant fuzz testing, static analysis with Slither, and exploit scenario simulation against a mainnet fork.

1.2 About Red Ghost Security

Red Ghost Security is a DeFi smart contract audit and exploit research firm specializing in oracle manipulation analysis, flash loan attack surface review, and zero-day vulnerability discovery. Our approach combines Foundry-based invariant fuzz testing with manual code review and live exploit simulation on mainnet forks.

Every vulnerability identified during our audits is validated as a working proof-of-concept against a mainnet fork, with precise economic impact measurement.

1.3 Audit Methodology

Manual Code Review

Line-by-line analysis of all in-scope smart contracts for logical flaws, semantic inconsistencies, and security vulnerabilities.

Static Analysis

Automated scanning using Slither and custom detectors to identify common vulnerability patterns.

Foundry Fuzz Testing

Invariant-based fuzz testing with Foundry to identify edge cases and state transition vulnerabilities.

Exploit Simulation

Each identified vulnerability is converted into a working PoC against a mainnet fork with precise economic impact measurement.

1.4 Disclaimer

This audit report is provided "as is" and does not constitute a guarantee or warranty that the audited smart contracts are free from all vulnerabilities. The audit was performed on a specific version of the codebase at a specific commit hash. Changes made after the audit are outside the scope of this report. Red Ghost Security shall not be held liable for any losses incurred as a result of using the audited smart contracts. This report is for informational purposes only and does not constitute financial or investment advice.

2. Findings Summary

2.1 Executive Summary

A total of 1 findings were identified during the audit of the BiFi Protocol protocol. These include 1 high.

2.2 Severity Breakdown

High	1
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ID	Title	Severity	Status
RGS-003	Cross-Chain Oracle Bridge Price Inconsiste	High	Open

RGS-003 -- HIGH

Cross-Chain Oracle Bridge Price Inconsistency

Description

Price data bridged across chains lacks synchronization validation. Token prices can differ between source and destination chains due to timing discrepancies in the oracle update mechanism.

Impact

An attacker can execute a flash loan on one chain to manipulate the source price, then exploit the stale bridge price on the destination chain before the oracle update propagates.

Exploit Scenario

1. Flash loan large position on Chain A
2. Manipulate spot price on a Uniswap V2 pair
3. Borrow against the manipulated price on Chain B before oracle update
4. Profit from the price discrepancy
5. Repay flash loan on Chain A

Recommendation

Implement a cross-chain oracle synchronization mechanism with minimum confirmation blocks. Add price deviation checks between chains.

4. Conclusion

The audit of the BiFi Protocol protocol identified 1 findings, including 0 critical-severity and 1 high-severity issues. The identified vulnerabilities have been communicated to the BiFi Protocol team along with remediation recommendations.

Red Ghost Security recommends that all identified issues be addressed before mainnet deployment. A follow-up audit is recommended to verify remediation.

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